

LIME and SHAP

An Overview of basic Explainable AI techniques



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Explainable AI: the basics

Explainable AI, as you have already seen, provides techniques to **understand better how machine learning algorithms work.**

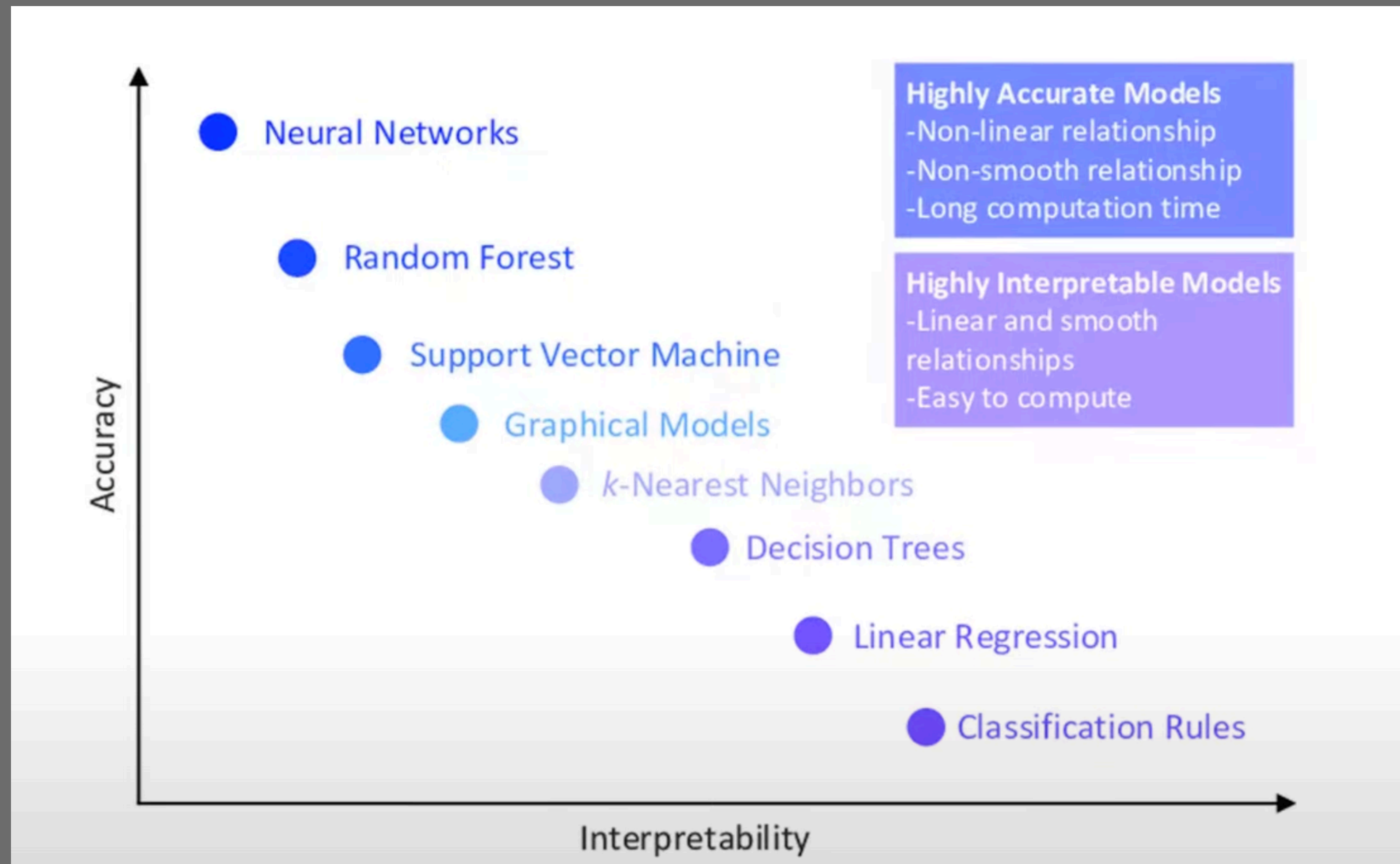
The demand for transparency in the decision-making process is a key driver for the adoption of Explainable AI.

Usually, when building a model starting from data, a trade-off appears that the more accurate the models **we make, the less interpretable they are.**

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Explainable AI: the basics

Explainable AI, or XAI, is a field of research that aims to help us **understand** better how machine learning models work.

The demand for explainable AI is growing rapidly, driven by the need for transparency and accountability in AI systems.

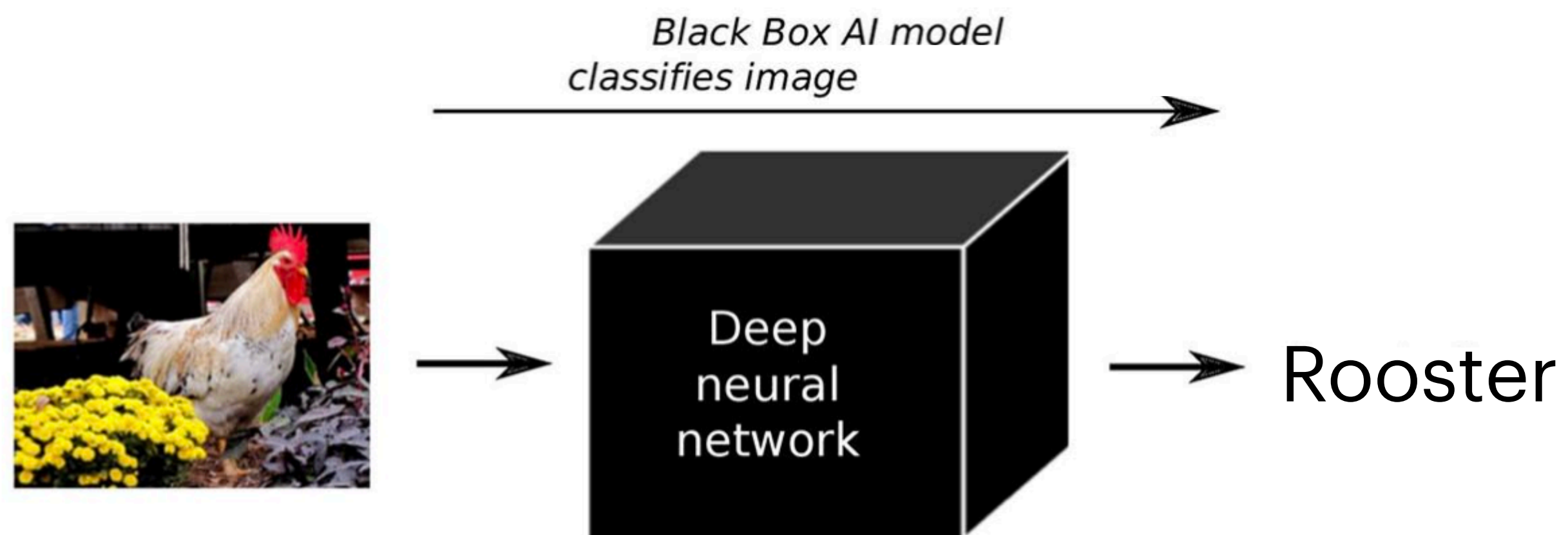
Usually, when we use machine learning models, we don't know **they are** making decisions. That's why we need XAI to help us understand what the model is doing.

But do we need XAI?
When and how it can be useful?
Let's make an example

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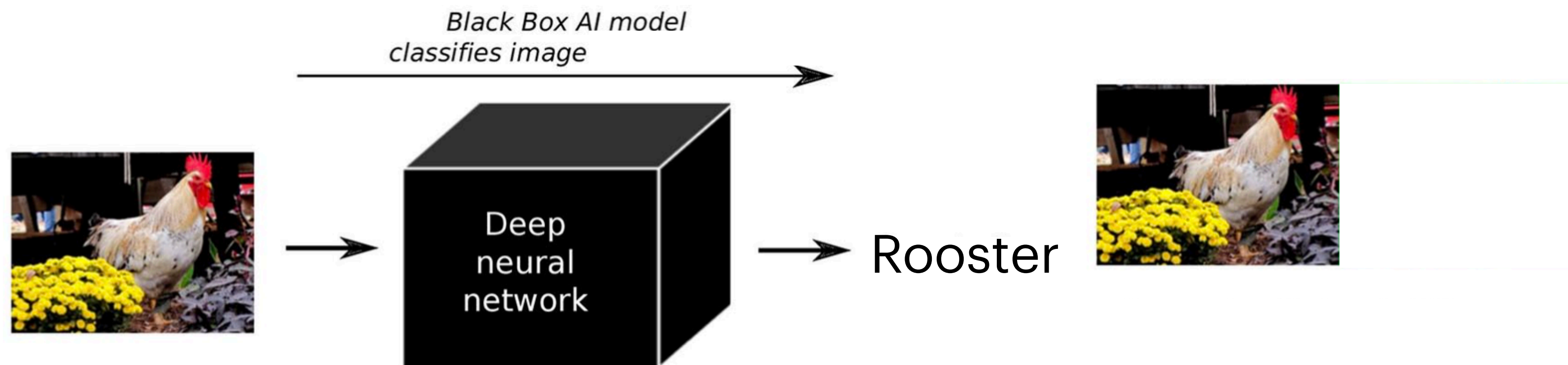
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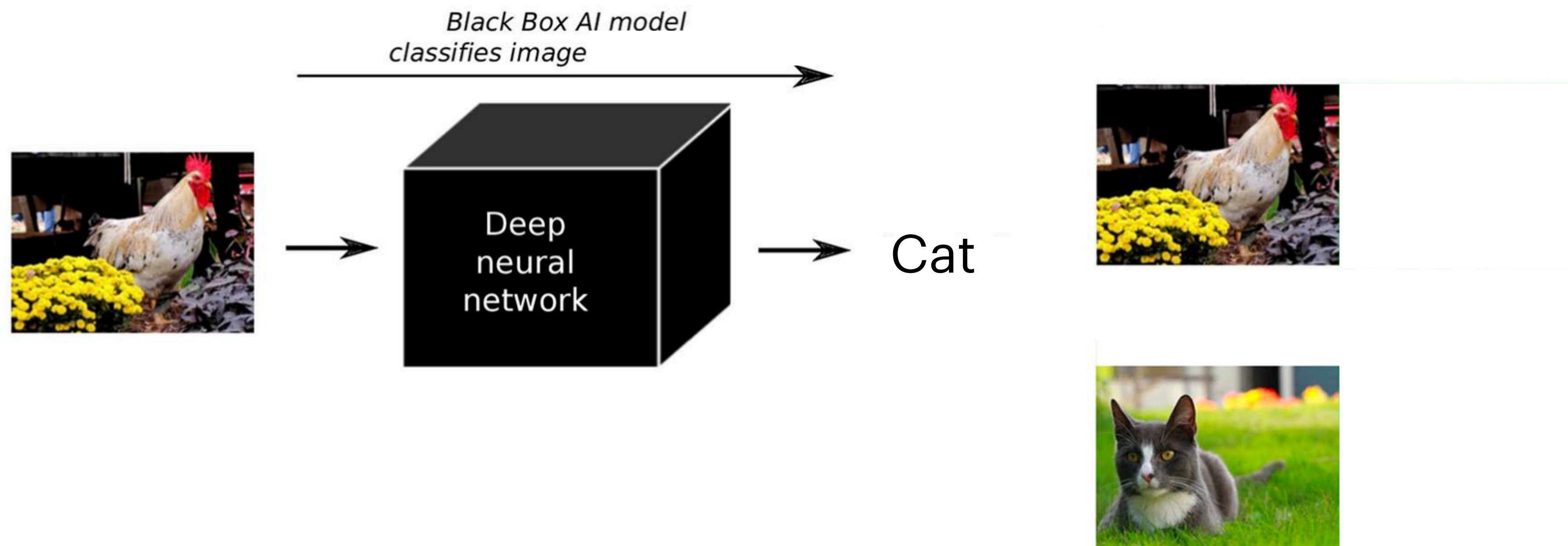
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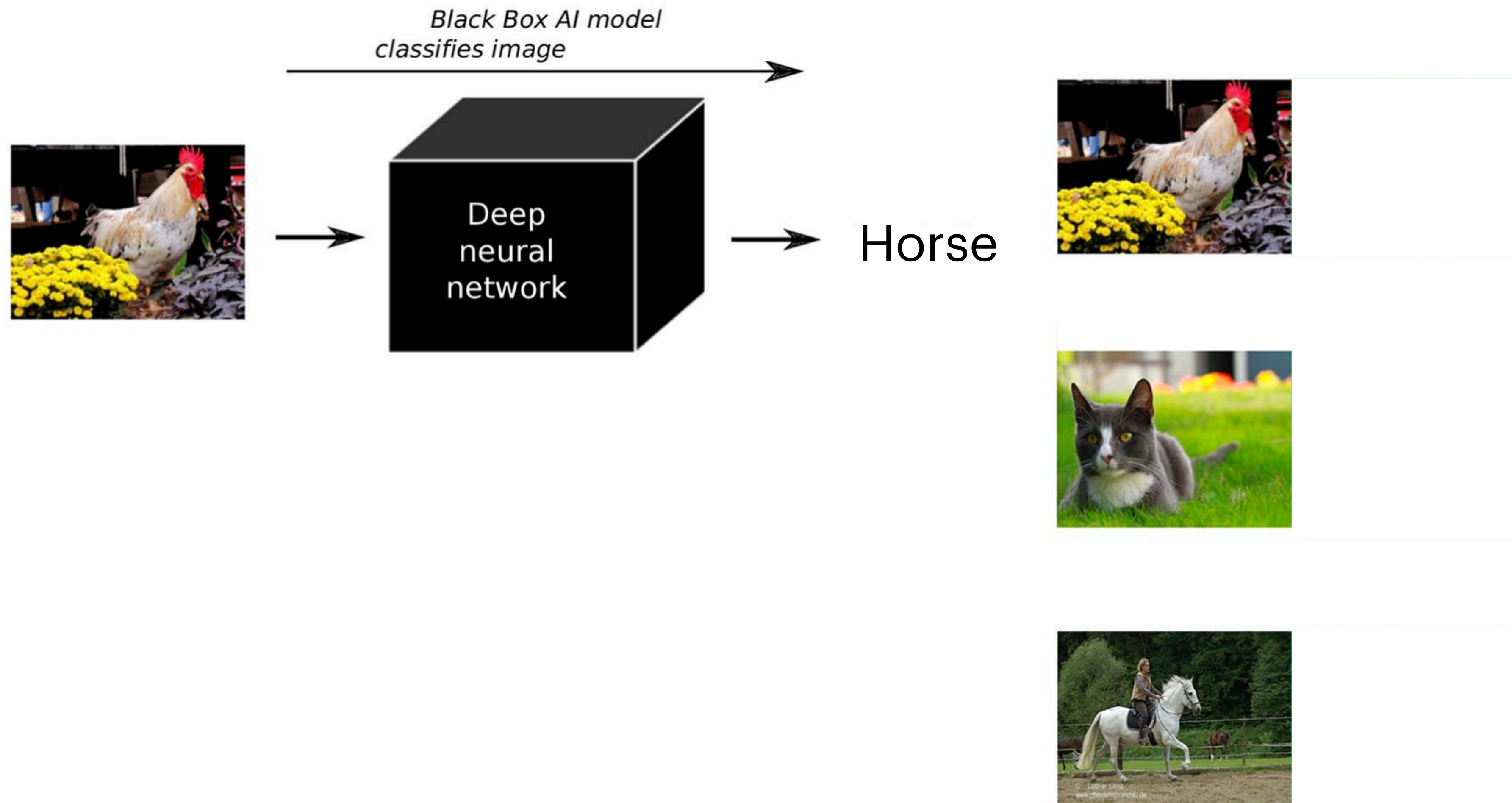
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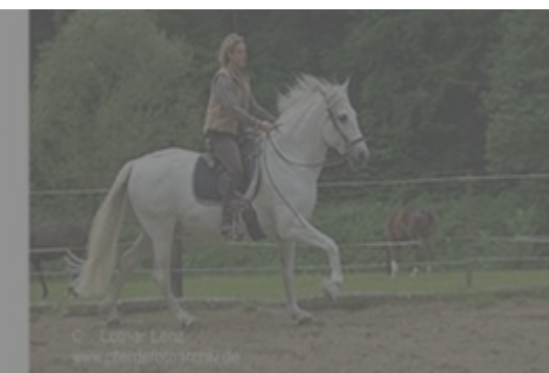
Context



Black Box AI model

So, we may think the model behaves correctly.

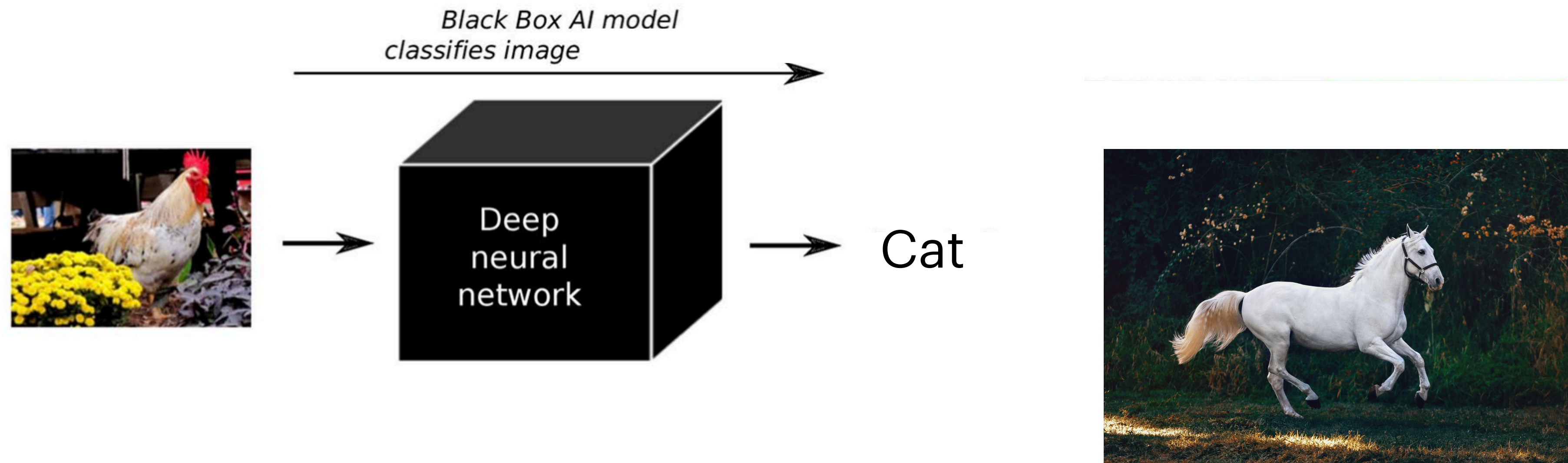
But, is it?



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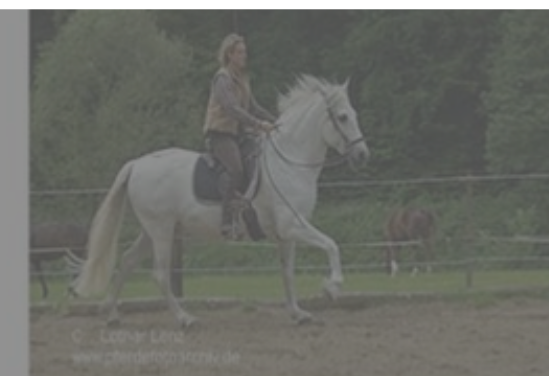
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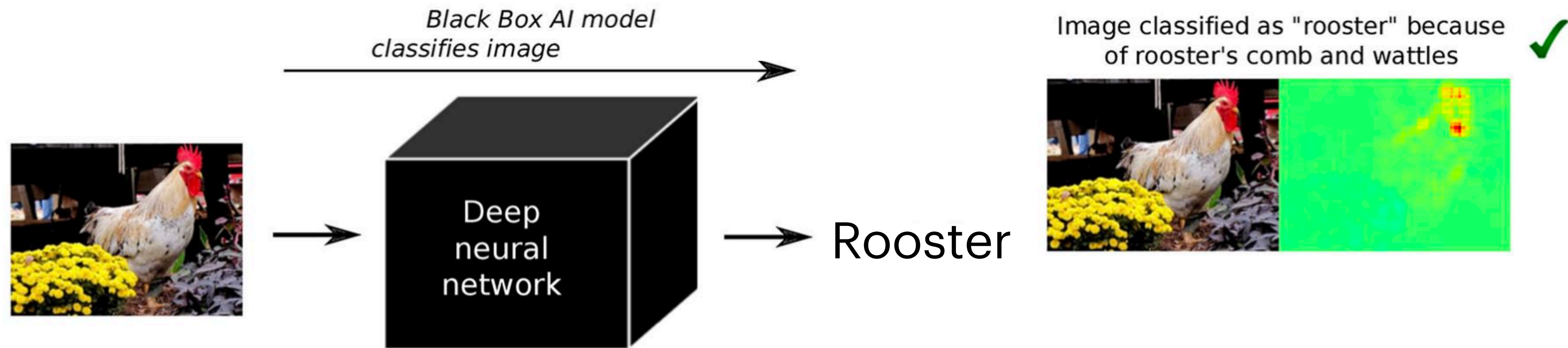
Let's apply Explainable AI to this model



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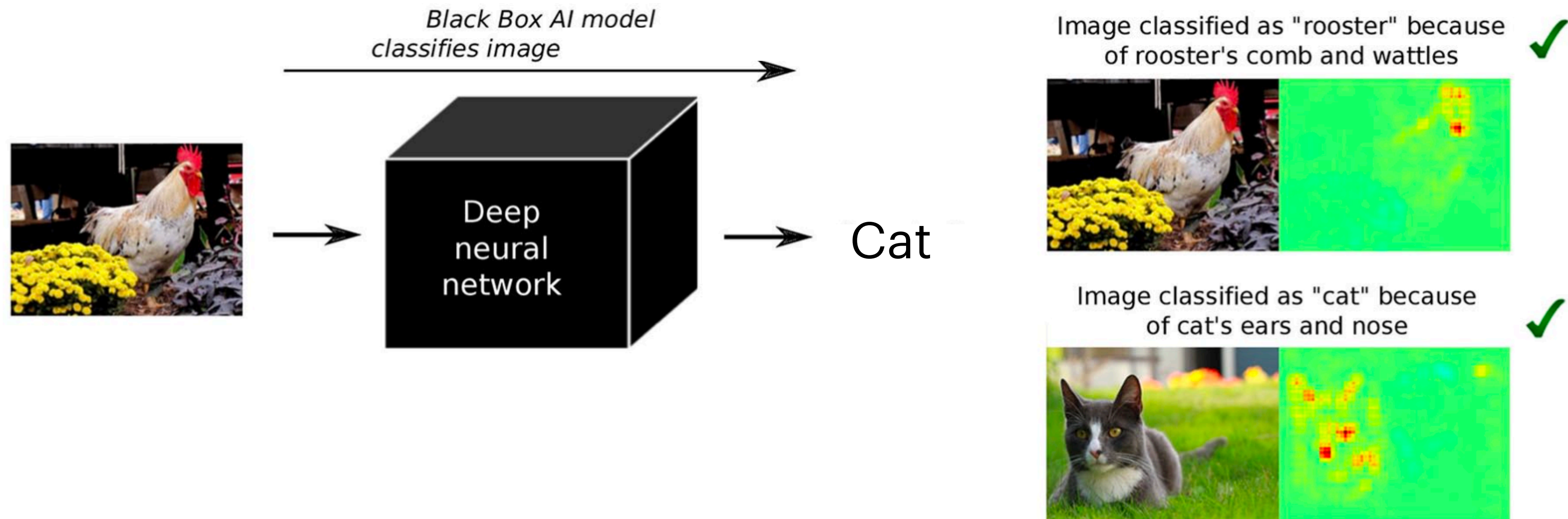
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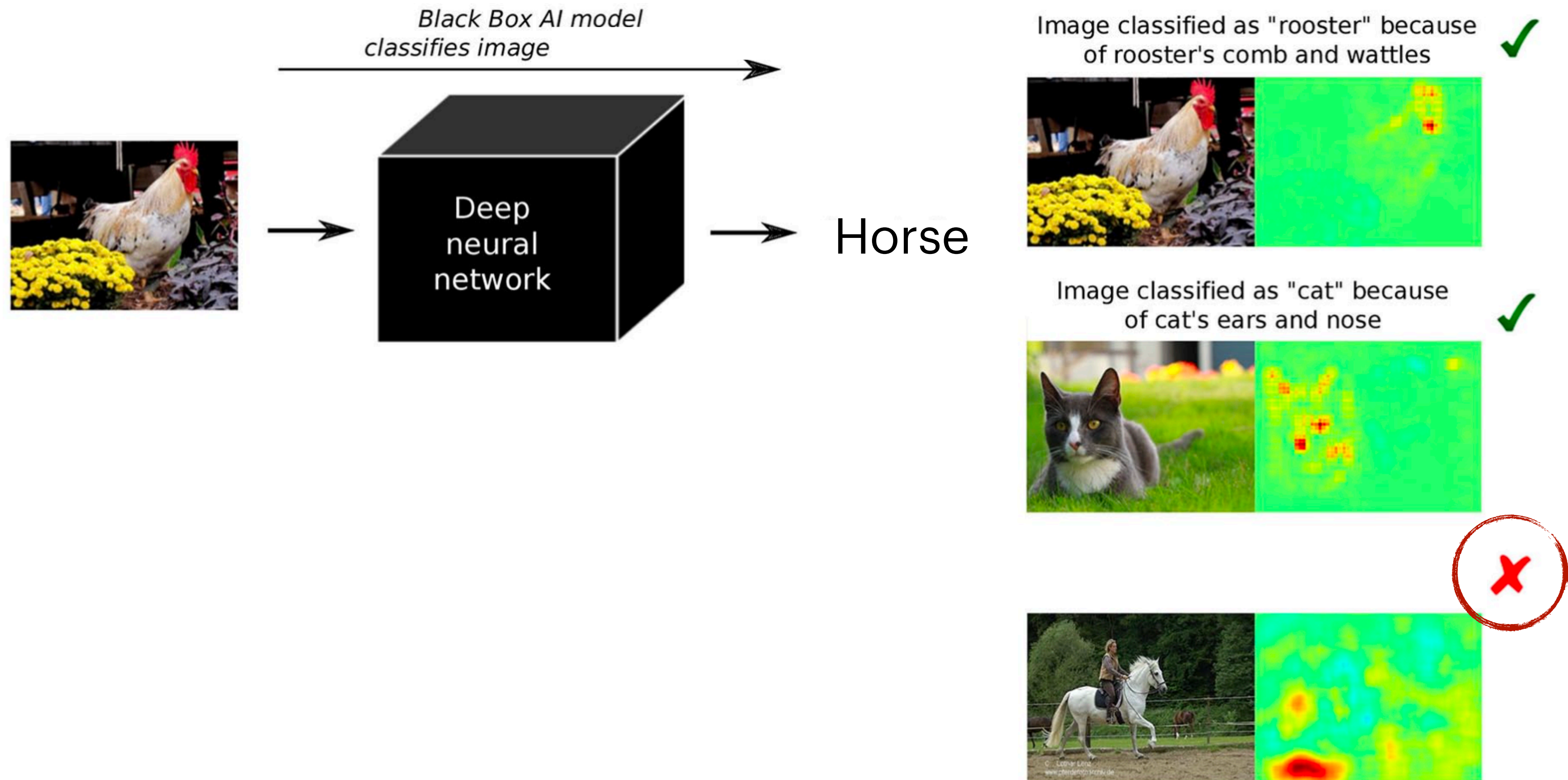
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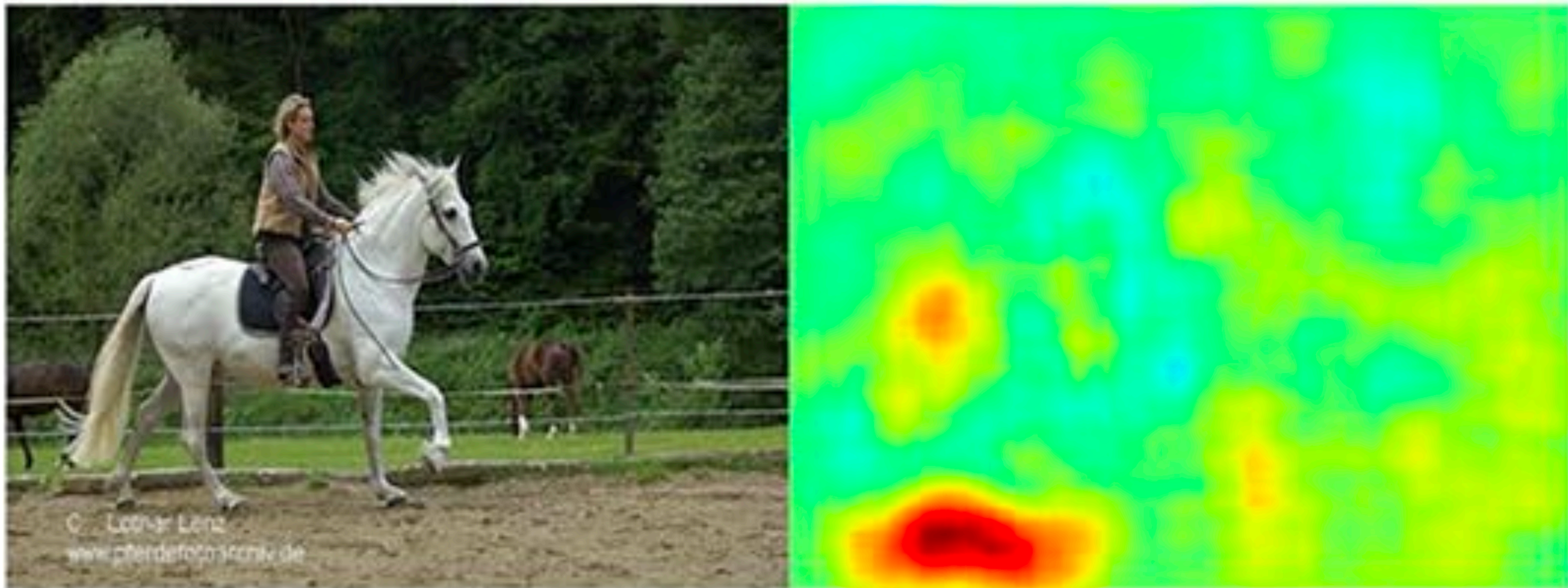


What Happened?

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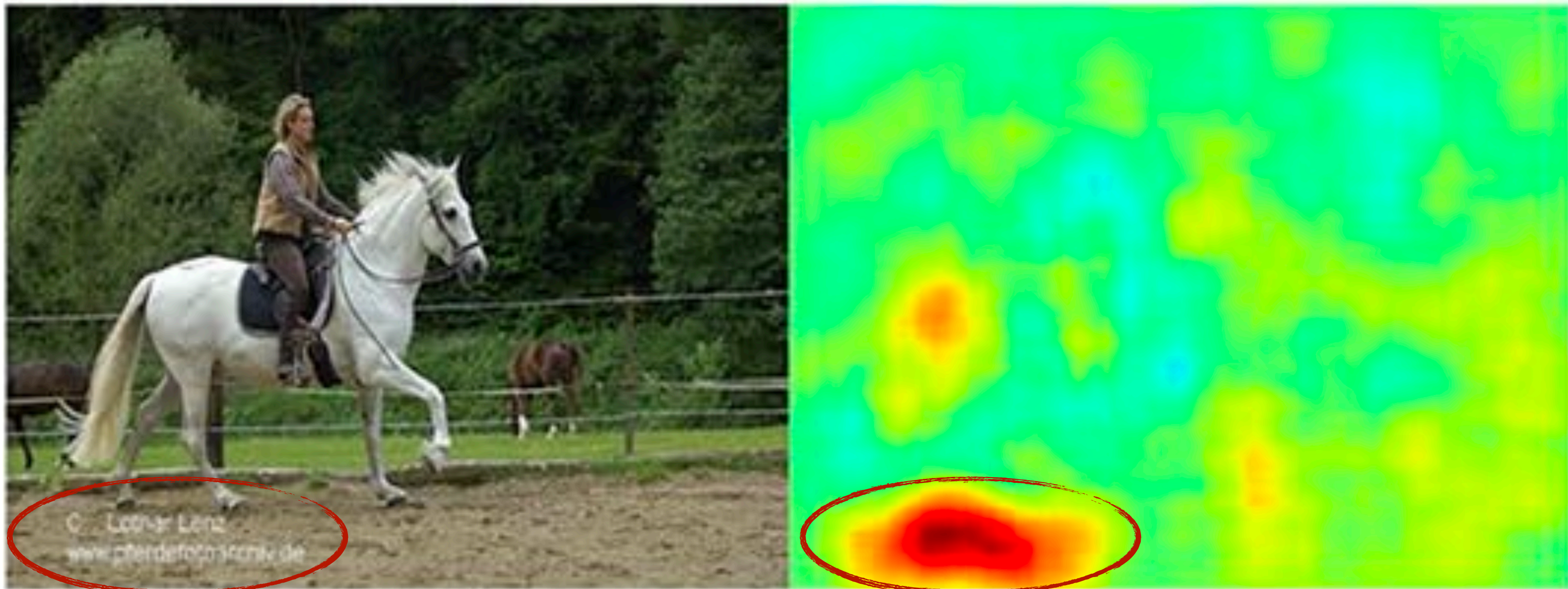
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The Copyright mark!

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Simply enough, there was an un-looked common characteristic on almost all the horses' photos that made the model think that the copyright mark on the photo of horses was indeed an important feature of a horse



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This experiment has been really done, and the complete version of it is in this paper.



LIME (Local Interpretable Model-agnostic Explanations)

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LIME aims to enhance the interpretability of complex models by generating **locally faithful and understandable explanations for individual predictions.**

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LIME tests what happens to the predictions when you give **variations of your data** into the machine learning model. It generates a new dataset consisting of **perturbed samples** and the corresponding predictions of the black box model.

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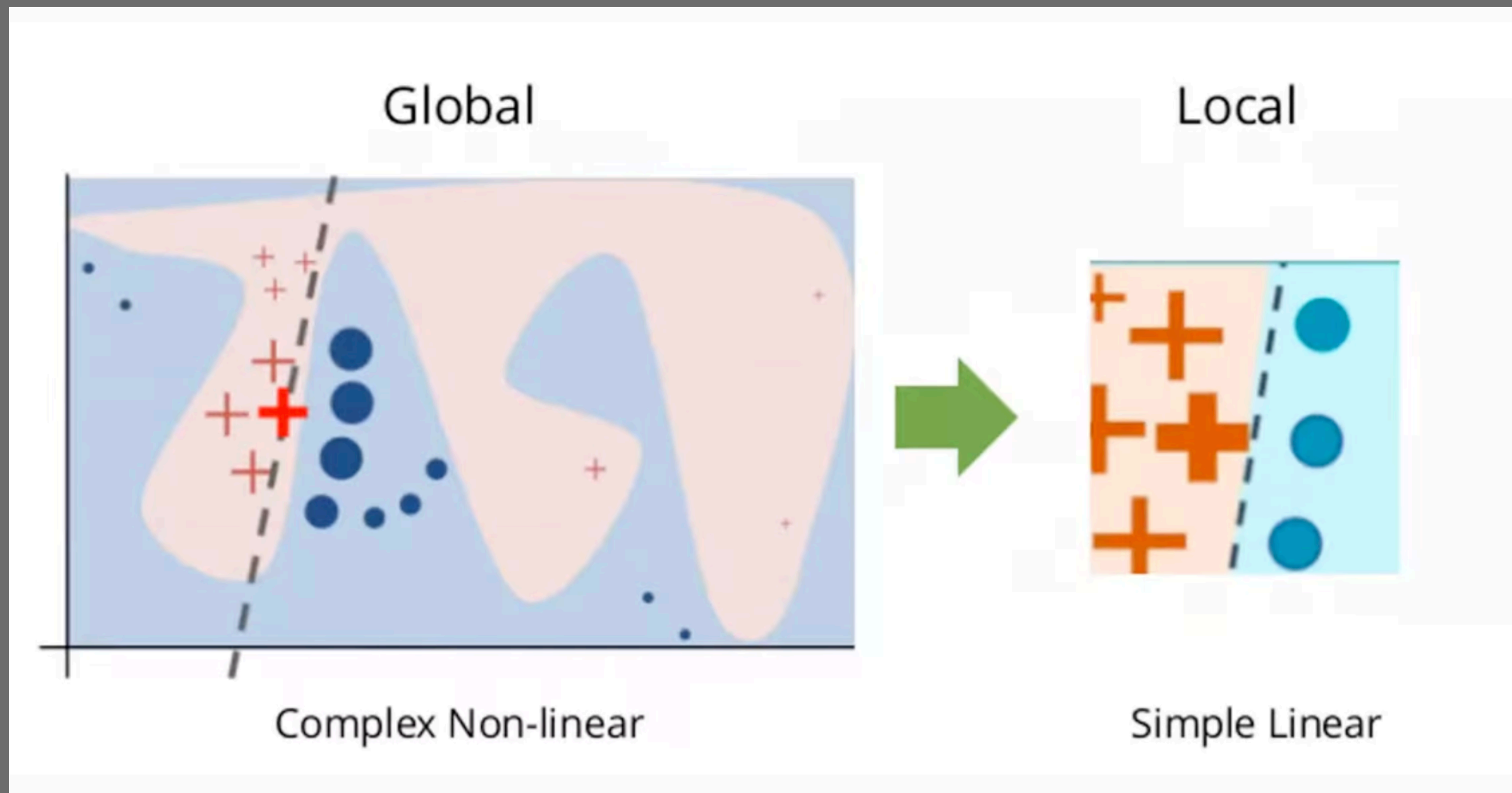
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LIME tests what happens to the predictions when you give **variations of your data** into the machine learning model. It generates a new dataset consisting of **perturbed samples** and the corresponding predictions of the black box model.

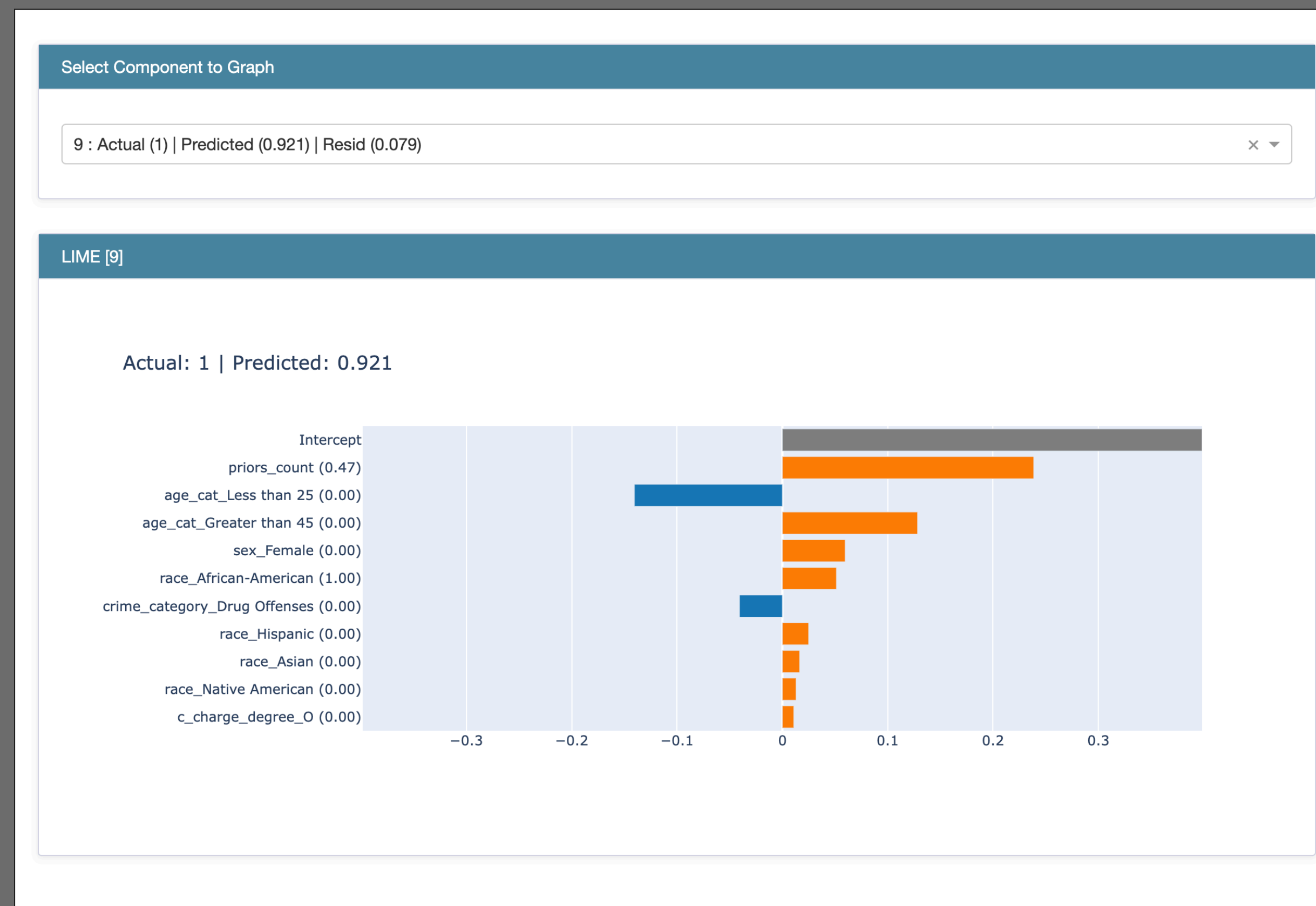
On this new dataset, LIME trains an interpretable model (**a local surrogate**) weighted by the **proximity of the sampled instances** to the instance of interest.

LIME (Local Interpretable Model-agnostic Explanations)



A graphical intuition of LIME's working

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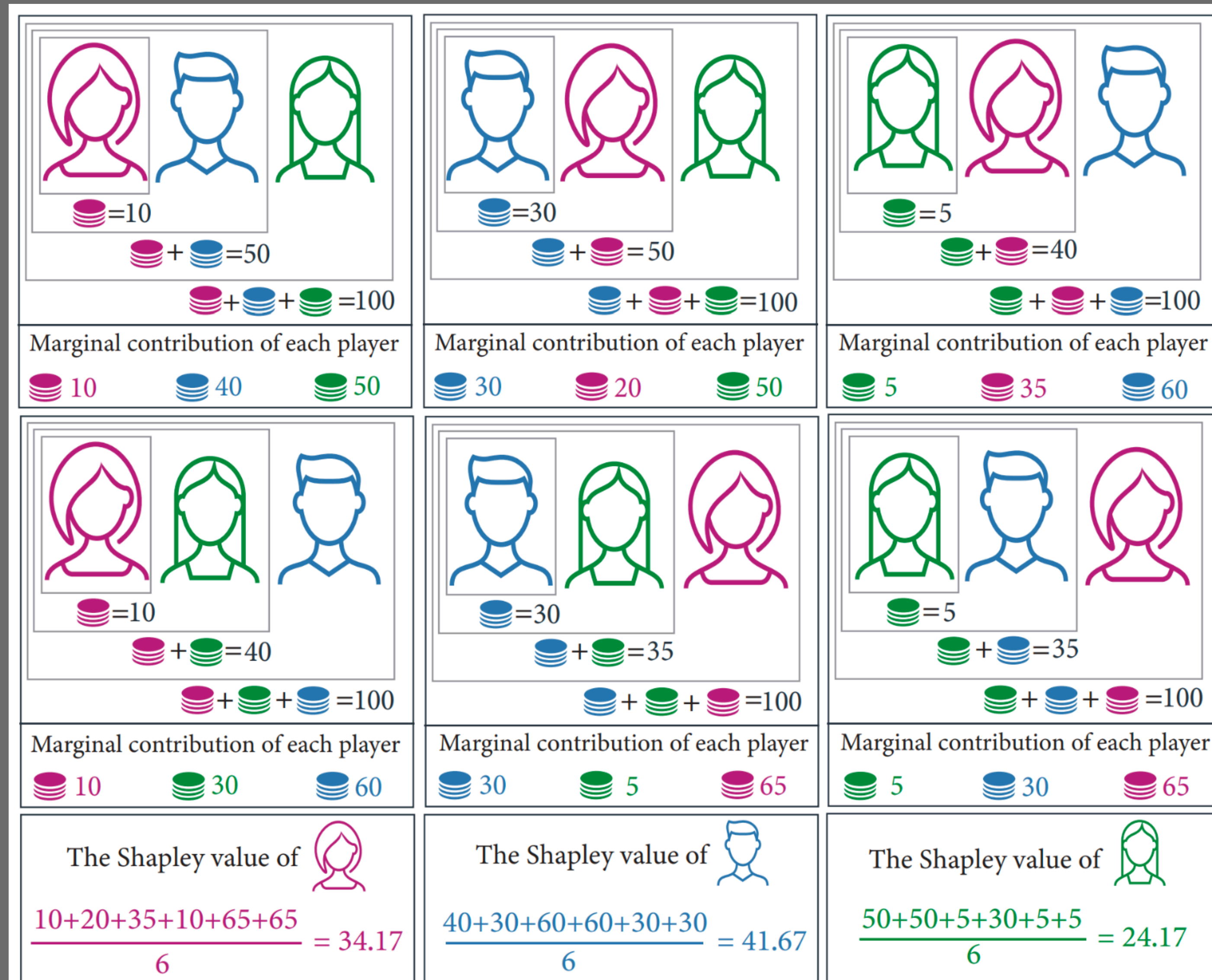
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SHAP computes these values by considering **all possible combinations of features**, ensuring a comprehensive and unbiased attribution.

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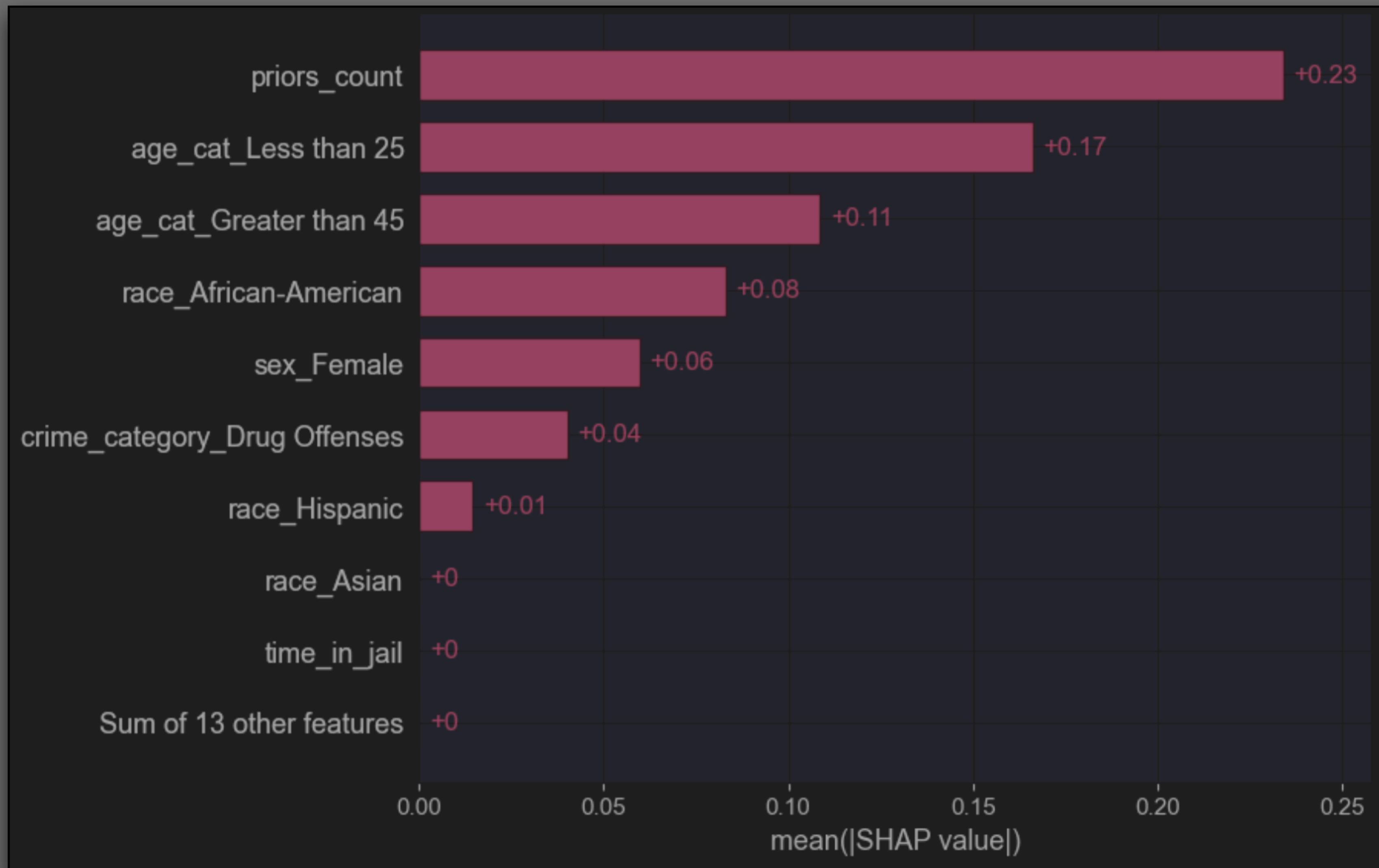
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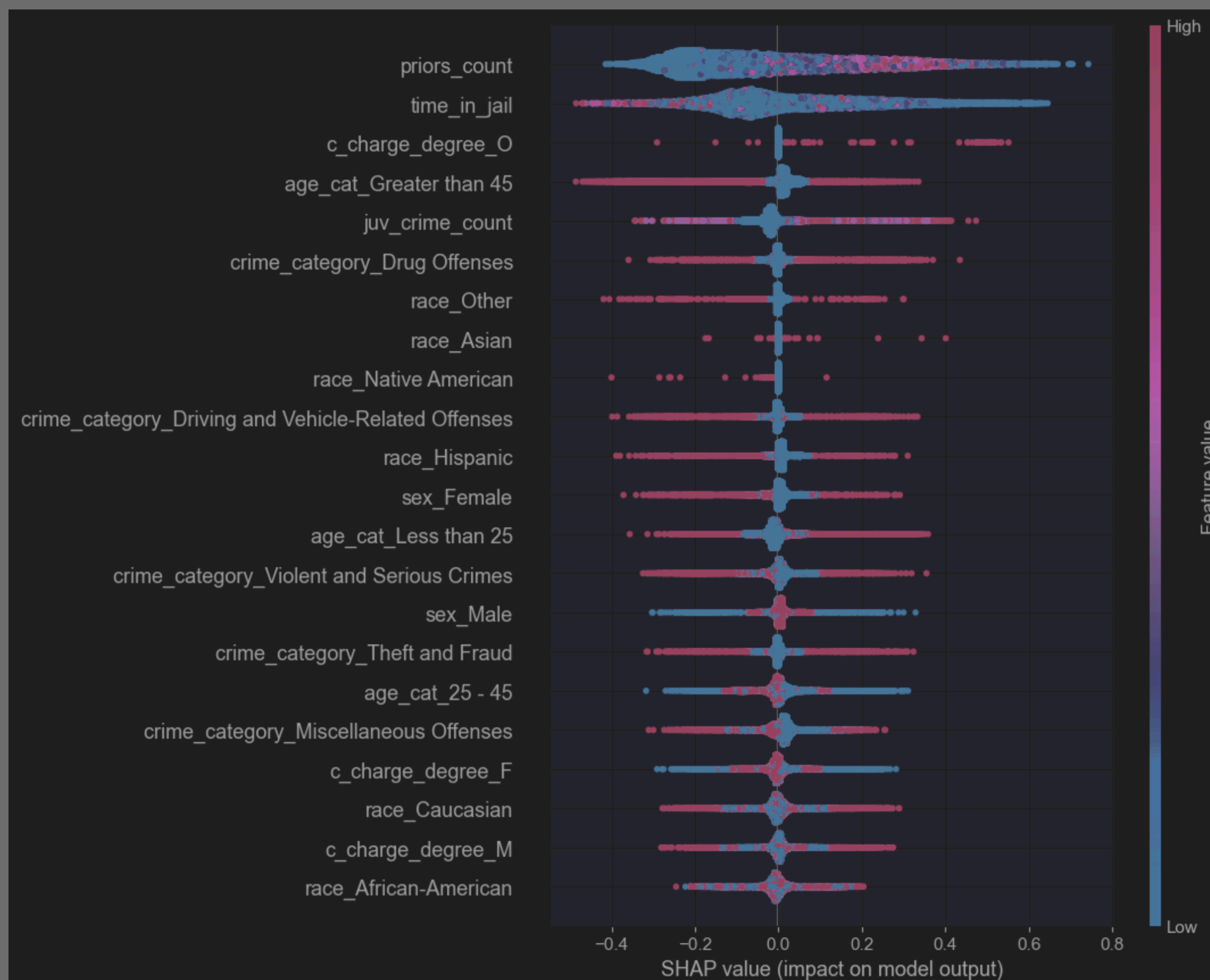
But, computing the Shapely Values is an **NP-Hard problem**, with time complexity $O(2^n)$

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If LIME and SHAP have so much in common (both of them performs local explanations for example), why one should prefer the one on the other?

	LIME	SHAP
Theory driven	Fails at being consistent. ✗	Supported by the Shapley values theory properties and consistency property. ✓
Time expensive	Time affordable. ✓	Computation of marginal contributions for all possible coalitions makes it time expensive. ✗
Require training data	Does not require the training set for fitting the surrogate model. ✓	Requires the training set for generating the background set that will be used to train the surrogate model. ✗